

Statistical Modeling – Quick Reference Notes

1. Hypothesis Testing

Hypothesis testing is used to check whether there is enough evidence to support a claim about data. We start with a **null hypothesis (H_0)** and an **alternative hypothesis (H_1)**. The p-value is compared with the significance level, usually **0.05**. If $p < 0.05$, we reject H_0 ; otherwise, we fail to reject H_0 .

2. p-value

The p-value tells us how much evidence we have against the null hypothesis. A small p-value means stronger evidence against H_0 . If $p < 0.05$, we reject H_0 . If $p \geq 0.05$, we fail to reject H_0 .

3. Shapiro-Wilk Test

The Shapiro-Wilk test checks whether data is approximately **normally distributed**. Its H_0 states that the data is normally distributed. If $p < 0.05$, we reject H_0 and conclude that the data is not normally distributed. In our project, it was used for the smoker and non-smoker groups.

4. Levene's Test

Levene's test checks whether two groups have **equal variances**. Its H_0 states that the variances of the groups are equal. If $p < 0.05$, we reject H_0 and conclude that the variances are different. It helps decide whether the equal-variance assumption is reasonable.

5. Mann-Whitney U Test

The Mann-Whitney U test compares **two independent groups**. It is useful when the data does not satisfy the normality assumption required for a t-test. In our project, it was used to compare medical charges between smokers and non-smokers. A small p-value indicates a significant difference between the groups.

6. One-Way ANOVA

One-Way ANOVA compares the means of **three or more groups**. Its H_0 states that all group means are equal, while H_1 states that at least one mean is different. In our project, we compared medical charges across the four regions. If $p < 0.05$, we reject H_0 .

7. Multiple Linear Regression

Multiple Linear Regression is used to predict a numerical target using **multiple independent variables**. In our project, medical charges are predicted using age, BMI, children, sex, smoker status, and region. It also helps us understand the relationship

between each predictor and medical charges. The other predictors are considered constant when interpreting one coefficient.

8. OLS

OLS (Ordinary Least Squares) is a method used to estimate the coefficients of a linear regression model. It finds the coefficients that make the overall squared prediction error as small as possible. The difference between an actual and predicted value is called a residual. In Python, we use `sm.OLS()` to fit the model.

9. Regression Coefficient

A regression coefficient tells us how the predicted target changes when one predictor increases by one unit, while the other predictors remain constant. A positive coefficient means the predicted target increases. A negative coefficient means it decreases. For example, our smoker coefficient shows a large increase in predicted medical charges for smokers compared with non-smokers.

10. R^2

R^2 tells us how much of the variation in the target variable is explained by the regression model. Its value usually lies between 0 and 1. Our model has **$R^2 \approx 0.751$** , meaning about **75.1% of the variation in medical charges** is explained by the predictors. A higher R^2 generally means the model explains more variation.

11. Adjusted R^2

Adjusted R^2 is similar to R^2 but also considers the **number of predictors** in the model. It prevents adding unnecessary variables from automatically making the model look better. Our adjusted R^2 is approximately **0.749**. It is especially useful when comparing regression models with different numbers of predictors.

12. Residual

A residual is the difference between the **actual value and predicted value**.

$$\text{Residual} = \text{Actual} - \text{Predicted}$$

For example, if the actual charge is ₹20,000 and the predicted charge is ₹18,000, the residual is ₹2,000. Residuals are used to check whether the regression model is performing appropriately.

13. Residual vs Fitted Plot

The Residual vs Fitted plot shows residuals against the model's predicted values. Ideally, the points should be randomly scattered around zero. A clear pattern may

indicate that the model is missing some information. A changing spread of residuals can indicate **heteroscedasticity**.

14. Q-Q Plot

A Q-Q plot is a visual method used to check whether residuals are approximately **normally distributed**. It compares the observed residual values with values expected from a normal distribution. If the points are close to the diagonal line, normality is more reasonable. Large deviations from the line suggest non-normality.

15. Jarque-Bera Test

The Jarque-Bera test checks whether the **residuals are normally distributed**. It mainly uses the skewness and kurtosis of the residuals. If $p < 0.05$, we reject the normality assumption. It is used as a regression diagnostic.

16. Breusch-Pagan Test

The Breusch-Pagan test checks whether the variance of the regression errors is **constant**. Its H_0 states that the error variance is constant, while H_1 states that it is not constant. If $p < 0.05$, we reject H_0 and conclude that heteroscedasticity is present. In our project, this test showed strong evidence of heteroscedasticity.

17. Multicollinearity

Multicollinearity occurs when two or more independent variables are **strongly related to each other**. It can make regression coefficients unstable and increase their standard errors. This makes it harder to understand the individual effect of predictors. We check for multicollinearity using VIF.

18. VIF

VIF (Variance Inflation Factor) is used to measure multicollinearity between predictors. A VIF close to 1 indicates very little multicollinearity. Values below 5 are generally considered acceptable, while higher values may indicate a problem. In our project, all VIF values were below 5.

19. HC3

HC3 (Heteroscedasticity-Consistent standard errors) is used when heteroscedasticity is present. It does not change the regression coefficients. Instead, it adjusts the standard errors, p-values, and confidence intervals to make the statistical inference more reliable. We used HC3 in our project because the Breusch-Pagan test detected heteroscedasticity.